

ESP32 Relay AC X2 Peripheral Connections

GPIO assignment

GPIO	ESPHome ID	Peripheral	Signal / function	Connection notes
GPIO2	button_minus	Push button	Decrease target temperature	Connect button between GPIO2 and GND; internal pull-up enabled
GPIO4	button_mode	Push button	Thermostat ON/OFF	Connect button between GPIO4 and GND; internal pull-up enabled
GPIO5	thermostat_display	TM1637 display	DIO	TM1637 data line
GPIO15	button_plus	Push button	Increase target temperature	Connect button between GPIO15 and GND; internal pull-up enabled
GPIO16	relay_1	On-board relay 1	Heating output	Internal ESPHome switch; controlled by the thermostat
GPIO17	relay_2	On-board relay 2	Auxiliary relay output	Exposed as Relay 2 in Home Assistant
GPIO18	thermostat_display	TM1637 display	CLK	TM1637 clock line
GPIO23	output_led	On-board LED	General-purpose LED	Exposed as LED in Home Assistant
GPIO25	I ² C bus	SHT3x sensor	SDA	I ² C data line
GPIO32	I ² C bus	SHT3x sensor	SCL	I ² C clock line

Peripheral wiring

SHT3x temperature and humidity sensor

SHT3x pin	ESP32 Relay AC X2 connection
VCC	3.3 V
GND	GND
SDA	GPIO25
SCL	GPIO32

- I²C address: 0x44
- I²C frequency: 50 kHz

TM1637 four-digit display

TM1637 pin	ESP32 Relay AC X2 connection
VCC	3.3 V or 5 V, depending on the module
GND	GND
CLK	GPIO18
DIO	GPIO5

Push buttons

Button	GPIO	Wiring
Temperature +	GPIO15	Button between GPIO15 and GND
Temperature -	GPIO2	Button between GPIO2 and GND
Mode	GPIO4	Button between GPIO4 and GND

The buttons use the ESP32 internal pull-up resistors and are therefore active LOW.

Relay outputs

Relay	GPIO	Purpose
Relay 1	GPIO16	Heating control
Relay 2	GPIO17	Auxiliary output / reserved

On-board LED

LED	GPIO	Purpose
On-board LED	GPIO23	General-purpose status LED